

An assessment of coefficient accuracy in linear regression models with spatially varying coefficients

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Received: 1 May 2006 / Accepted: 13 July 2006 / Published online: 18 January 2007
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Abstract The realization in the statistical and geographical sciences that a relationship between an explanatory variable and a response variable in a linear regression model is not always constant across a study area has led to the development of regression models that allow for spatially varying coefficients. Two competing models of this type are geographically weighted regression (GWR) and Bayesian regression models with spatially varying coefficient processes (SVCP). In the application of these spatially varying coefficient models, marginal inference on the regression coefficient spatial processes is typically of primary interest. In light of this fact, there is a need to assess the validity of such marginal inferences, since these inferences may be misleading in the presence of explanatory variable collinearity. In this paper, we present the results of a simulation study designed to evaluate the sensitivity of the spatially varying coefficients in the competing models to various levels of collinearity. The simulation study results show that the Bayesian regression model produces more accurate inferences on the regression coefficients than does GWR. In addition, the Bayesian regression model is overall fairly robust in terms of marginal coefficient inference to moderate levels of collinearity, and degrades less substantially than GWR with strong collinearity.

Keywords Bayesian regression · Geographically weighted regression · MCMC · Simulation study · Spatial statistics · Collinearity

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1 Introduction

The realization in the statistical and geographical sciences that a relationship between an explanatory variable and a response variable in a regression model is not necessarily constant across a study area has lead to the development of linear regression models that allow for spatially varying coefficients. In the field of geography, the geographically weighted regression (GWR) model has become an increasingly popular tool in recent years (e.g., Fotheringham et al. 2002). GWR is similar in spirit and methodology to local linear regression models that were popularized in the statistics literature (see Loader 1999), except that in GWR weights are determined by a spatial kernel function instead of a kernel function in the variable space. There are numerous examples of applying GWR to real world data in the literature (Huang and Leung 2002; Longley and Tobón 2004; Nakaya 2001). Wheeler and Tiefelsdorf (2005) analyze this method with regard to regression coefficient correlation and find that the dependence in the coefficients can be problematic in some cases.

In the statistics literature, Bayesian regression models with spatially varying coefficient processes have been introduced to model non-constant linear relationships between variables (Gelfand et al. 2003). For convenience, we will refer to this type of model as SVCP for spatially varying coefficient process model. As with GWR, the motivation for these models is that, in certain applications, regression coefficients may vary at the local or regional level. However, instead of fitting spatially local regression models, in the SVCP framework, the spatial varying coefficients are modeled as a multivariate spatial process. Such an approach fits naturally into the Bayesian paradigm, where parameters are treated as unknown random quantities. The SVCP and GWR models are similar in nature due to their same focus on the interpretation of the regression coefficient patterns and the overall fit to the response variable. The SVCP model differs from GWR in that it is a single statistical model specified in a hierarchical manner. In contrast, GWR is an ensemble of local spatial regression models, each fit separately. Therefore, with the GWR model, there is an explicit disconnect between regression model coefficients locally; however, with the SVCP model, the dependence between regression coefficients is defined globally.

One topic that is understudied in the literature related to both types of spatial varying coefficient regression models is the validity of marginal inference on the regression coefficients, particularly in the presence of collinearity in the explanatory variables. The inherent assumption in the published papers that apply these models to data is that the regression coefficients are free of artifacts and strong dependence and, therefore, useful for marginal interpretation. This is, of course, an important and questionable assumption. In fact, there is no published work that assesses the validity of inferences derived from the GWR and SVCP models. In addition, it is well known that in linear regression models, strong collinearity in the explanatory variables can increase the variance of the estimated regression coefficients. Wheeler and

Tiefelsdorf (2005) and Wheeler (2006) show that collinearity can degrade coefficient precision in GWR and lead to counter-intuitive signs for some regression coefficients. Hence, in this paper we investigate coverage probabilities and accuracy of estimated regression coefficients from the GWR and SVCP models through simulation study, where the ‘true’ values of the regression coefficients are known. We evaluate the coverage and accuracy of the regression coefficients both in the absence of collinearity and in the presence of various levels of collinearity using coverage probabilities and a measure of deviation from the true values of the coefficients used to simulate the data. We compare the GWR and SVCP models based on these measures.

2 Overview of spatially varying coefficient models

2.1 Geographically weighted regression

This section reviews the key equations used in fitting a GWR model. The reader is referred to Fotheringham et al. (2002) for a more detailed introduction to the GWR framework. In GWR, the data are usually mean measures of aggregate data at fixed points with spatial coordinates; for example, see the Jiangsu province data in Huang and Leung (2002) and the numerous examples in Fotheringham et al. (2002). The spatial coordinates of the data are used to calculate distances that are used in a kernel function to determine weights of spatial dependence between observations. Typically, a separate regression model is fitted at each point location in the dataset, called model calibration locations. For each calibration location, $s = 1, \dots, n$, the GWR model at location s is

$$y(s) = \mathbf{X}(s)\boldsymbol{\beta}(s) + \boldsymbol{\varepsilon}(s), \quad (1)$$

where $y(s)$ is the dependent variable at location s , $\boldsymbol{\beta}(s)$ is the column vector of regression coefficients at location s , $\mathbf{X}(s)$ is the row vector of explanatory variables at location s , and $\boldsymbol{\varepsilon}(s)$ is the random error at location s . The vector of estimated regression coefficients at location s is

$$\hat{\boldsymbol{\beta}}(s) = [\mathbf{X}^T \cdot \mathbf{W}(s) \cdot \mathbf{X}]^{-1} \mathbf{X}^T \cdot \mathbf{W}(s) \cdot \mathbf{y}, \quad (2)$$

where $\mathbf{X} = [\mathbf{X}(1); \mathbf{X}(2); \dots; \mathbf{X}(n)]^T$ is the design matrix of explanatory variables, which typically includes a column of 1's for the intercept; $\mathbf{W}(s) = \text{diag}[w_1(s), \dots, w_n(s)]$ is the diagonal weights matrix that is calculated for each calibration location s ; $\mathbf{y}$ is the $n \times 1$ vector of dependent variables; and $\hat{\boldsymbol{\beta}}(s) = (\hat{\beta}_{s0}, \hat{\beta}_{s1}, \dots, \hat{\beta}_{sp})^T$ is the vector of $p + 1$ local regression coefficients at location s for p explanatory variables and an intercept.

The weights matrix, $\mathbf{W}(s)$, is calculated from a kernel function that places more weight on observations that are closer to the calibration location s . There are numerous choices for the kernel function, including the bi-square

nearest neighbor function, the exponential function, and the Gaussian function. We use the exponential function for the kernel in this paper to match the spatial dependence function used in the SVCP model, although one could also use another type in the Bayesian model. The weights from the exponential kernel function are calculated as

$$w_j(s) = \exp(-d_{sj}/\gamma), \quad (3)$$

where d_{sj} is the distance between the calibration location s and location j , and γ is the kernel bandwidth parameter.

To fit the GWR model, the kernel bandwidth is first estimated by cross-validation across all the calibration locations. Cross-validation is an iterative process that finds the kernel bandwidth with the lowest prediction error of all the $y(s)$. For each location s , it removes data for observation s in the model calibration at location s and predicts the response $y(s)$ using the other data points and the kernel weights associated with the current bandwidth. Next, the weights are calculated at each calibration location using the kernel function. Finally, the regression coefficients are estimated at each calibration location along with the response estimates by the expression $\hat{y}(s) = \mathbf{X}(s)\hat{\boldsymbol{\beta}}(s)$.

To measure the correlation in the sets of estimated regression coefficients, one can easily calculate their correlation coefficients using

$$C_{kl} \equiv \text{Corr}\left(\left\{\hat{\beta}_{1k}, \dots, \hat{\beta}_{nk}\right\}, \left\{\hat{\beta}_{1l}, \dots, \hat{\beta}_{nl}\right\}\right) = \frac{\sum_{s=1}^n (\hat{\beta}_{sk} - \bar{\hat{\beta}}_k) \cdot (\hat{\beta}_{sl} - \bar{\hat{\beta}}_l)}{\sqrt{\sum_{s=1}^n (\hat{\beta}_{sk} - \bar{\hat{\beta}}_k)^2 \cdot \sum_{s=1}^n (\hat{\beta}_{sl} - \bar{\hat{\beta}}_l)^2}}, \quad (4)$$

where $\bar{\hat{\beta}}_k = 1/n \cdot \sum_{s=1}^n \hat{\beta}_{sk}$. This correlation is subsequently called the *overall correlation coefficient* of two sets of local regression coefficients and is indicated C_{kl} for the correlation between variables k and l over all locations in the study area. Fotheringham et al. (2002) also present an expression for the “local regression coefficient covariance” at each location. This expression is not technically equal to the covariance of the GWR coefficients since GWR is not a formal statistical model with kernel weights that are part of the error structure. We will, however, use this expression as an exploratory tool for examining the correlation in the local coefficients. The equation is

$$\text{Cov}[\hat{\boldsymbol{\beta}}(s)] = \sigma^2 \cdot [\mathbf{X}^T \cdot \mathbf{W}(s) \cdot \mathbf{X}]^{-1} \cdot \mathbf{X}^T \cdot \mathbf{W}^2(s) \cdot \mathbf{X} \cdot [\mathbf{X}^T \cdot \mathbf{W}(s) \cdot \mathbf{X}]^{-1}. \quad (5)$$

A local parameter correlation matrix can be calculated from the local covariance matrix as

$$C(s) = \text{diag}^{-\frac{1}{2}}\{\text{Cov}[\hat{\boldsymbol{\beta}}(s)]\} \cdot \text{Cov}[\hat{\boldsymbol{\beta}}(s)] \cdot \text{diag}^{-\frac{1}{2}}\{\text{Cov}[\hat{\boldsymbol{\beta}}(s)]\}, \quad (6)$$

where the $\text{diag}\{\cdot\}$ operator extracts the diagonal elements of its argument. These two equations used for the local coefficient covariance and correlation

are only approximate equations because the kernel weights are calculated from the data using cross-validation before the regression coefficients are estimated from the data. The weights are inherently a function of $\mathbf{y}$, as are the regression coefficients, and the correct expression for the coefficient covariance is non-linear. We will not attempt to derive the exact covariance formula here and instead will use the approximate formula throughout the discussion. The correlation between coefficients for variables k and l at location s is indicated C_{kl}^s and comes from the (k, l) element of the $C(s)$ correlation matrix. Subsequently, we refer to these correlations as the *local coefficient correlations* at model calibration location s . The pair-wise coefficient correlations can be mapped at each calibration location for each pair of estimated coefficients.

2.2 Bayesian spatially varying coefficient process model

The Bayesian SVCP regression model is specified in a hierarchical manner. The distribution of the data is specified conditional on unknown parameters, whose distribution is in turn specified conditional on other parameters. Following Gelfand et al. (2003), the SVCP model is

$$[\mathbf{Y}|\boldsymbol{\beta}, \tau^2] = N(\mathbf{X}^T \boldsymbol{\beta}, \tau^2 \mathbf{I}), \quad (7)$$

where the bracket notation $[A|B]$ denotes the distribution of A conditional on B . $\mathbf{Y}$ is a vector of responses assumed to be Gaussian conditional on the parameters $\boldsymbol{\beta}$ and τ^2 ; $\boldsymbol{\beta}$ is a $np \times 1$ vector of regression coefficient parameters; and $\mathbf{X}^T$ is the $n \times np$ block diagonal matrix of covariates where each row contains a row from the $n \times p$ design matrix $\mathbf{X}^*$, along with zeros in the appropriate places (the covariates from $\mathbf{X}^*$ are shifted p places in each subsequent row in $\mathbf{X}^T$); $\mathbf{I}$ is the $n \times n$ identity matrix; and τ^2 is the error variance.

In the second stage of the hierarchical model, the prior distribution for the regression coefficient parameters is specified as

$$[\boldsymbol{\beta}|\boldsymbol{\mu}_\beta, \Sigma_\beta] = N(\mathbf{1}_{n \times 1} \otimes \boldsymbol{\mu}_\beta, \Sigma_\beta). \quad (8)$$

The vector $\boldsymbol{\mu}_\beta = (\mu_{\beta_0}, \dots, \mu_{\beta_p})^T$ contains the means of the regression coefficients corresponding to each of the p explanatory variables. The symbol $\otimes$ denotes the Kronecker product operator, which is the product of every element in $\mathbf{1}_{n \times 1}$ by $\boldsymbol{\mu}_\beta$. The prior on the regression coefficients in the SVCP model takes into account the possible spatial dependence in the coefficients through the covariance, Σ_β . For $\boldsymbol{\beta}_p = [\beta_{p1}, \dots, \beta_{pn}]$, we can assume a priori that each β_p follows an areal unit model (e.g., the conditional autoregressive model or simultaneous autoregressive model; see Banerjee et al. 2004) or specify the prior on $\boldsymbol{\beta}_p$ using a geostatistical approach, where a parametric distance-based covariance function is specified. We focus on a geostatistical prior specification of the regression coefficients and assume an exponential spatial dependence function. The prior covariance matrix for the p different types of $\boldsymbol{\beta}$'s at each of n locations, Σ_β , can have either a separable or nonseparable form. The

separable form has two distinct components, one for the spatial dependence in the regression coefficients and one for the within site dependence between coefficients of the same type. Following Gelfand et al. (2003), we assume a separable covariance matrix for $\boldsymbol{\beta}$ of the form

$$\Sigma_{\boldsymbol{\beta}} = \mathbf{H}(\phi) \otimes \mathbf{T}, \quad (9)$$

where $\mathbf{T}$ is a positive-definite $p \times p$ matrix for the covariance of the regression coefficients at any spatial location, $\mathbf{H}(\phi)$ is the $n \times n$ correlation matrix that captures the spatial association between the n locations, and ϕ is an unknown spatial dependence parameter. In the prior specification for $\boldsymbol{\beta}$ (Eq. 9), the Kronecker product results in a $np \times np$ positive definite covariance matrix, since $\mathbf{H}(\phi)$ and $\mathbf{T}$ are both positive definite. The elements of the correlation matrix $\mathbf{H}(\phi)$, $H(\phi)_{ij} = \rho(s_i - s_j; \phi)$, are calculated from the exponential function $\rho(d; \phi) = \exp(-d/\phi)$.

With the separable cross-covariance function, each of the p coefficients represented in the covariance is assumed to have the same spatial dependence structure. This aligns with the assumption of equal spatial ranges for each explanatory variable in the GWR model, and justifies our use of the separable form. The separable cross-covariance form also has the property that the covariance between $\boldsymbol{\beta}$'s of the same type is constant across space. While a separable assumption is restrictive, one advantage to the separable covariance is that it is more convenient computationally than a nonseparable one and reduces the number of operations needed for matrix inversion in simulating from the posterior distribution of the parameters. A nonseparable form would allow for distinct spatial ranges for each coefficient process, and is therefore more flexible. An example of the use of a nonseparable covariance matrix in a Bayesian regression model is Banerjee and Johnson (2005), who use a linear model of coregionalization to specify the prior on $\boldsymbol{\beta}$ (see also Banerjee et al. 2004; Gelfand et al. 2004 for discussions of models of coregionalization).

The specification of the Bayesian SVCP model in Eqs. 7, 8, and 9 is complete with the specification of the prior distributions of the parameters. The prior for the error variance is inverse gamma with hyperparameters a and b , $[\tau^2] \sim IG(a, b)$. Hyperparameters control the mean and variance of the prior distribution of the parameter. The prior for the coefficient means is normal with hyperparameters $\boldsymbol{\mu}$ and σ^2 , $[\boldsymbol{\mu}_{\boldsymbol{\beta}}] \sim N(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$. The prior for the covariance matrix $\mathbf{T}$ is inverse Wishart with hyperparameters ν and $\boldsymbol{\Omega}$, $[\mathbf{T}] \sim IW_{\nu}(\boldsymbol{\Omega}^{-1})$. These priors are conjugate priors, and are used for computational convenience. The prior for the spatial dependence parameter ϕ is gamma with hyperparameters α and λ , $[\phi] \sim G(\alpha, \lambda)$. The parameterizations of the gamma distribution and inverse Wishart distribution used in this manuscript are listed in the Appendix.

Inference on the parameters in the SVCP model is based on the posterior distribution $[\boldsymbol{\theta}|\mathbf{y}]$ of the parameters $\boldsymbol{\theta} = (\boldsymbol{\beta}, \tau^2, \boldsymbol{\mu}_{\boldsymbol{\beta}}, \phi, \mathbf{T})$, which can be obtained using Bayes Theorem:

$$[\boldsymbol{\theta}|\mathbf{y}] \propto [\mathbf{y}|\boldsymbol{\theta}][\boldsymbol{\theta}]. \quad (10)$$

In other words, the posterior distribution for the parameters $\boldsymbol{\theta}$, conditional on the data, is proportional to the likelihood of the data $[\mathbf{y}|\boldsymbol{\theta}]$, also written as $p(\mathbf{y}|\boldsymbol{\theta})$, and the prior $[\boldsymbol{\theta}]$, also written as $p(\boldsymbol{\theta})$, for all the parameters. In most complex Bayesian models, it usually is not possible to analytically derive the posterior distribution of model parameters. Instead, simulation-based inference tools, such as Markov chain Monte Carlo (MCMC) methods, can be used to approximate the posterior distribution. MCMC algorithms simulate a Markov chain that has for its stationary distribution the target posterior distribution. Thus, the sample path of the Markov chain produces samples from the joint posterior distribution of the model parameters. The algorithm must be run for a sufficient time so that, after a number of ‘burn-in’ iterations, the algorithm converges and the sample path of the Markov chain can be taken to be samples from the posterior distribution of the unknown parameters. The samples from the chain after the ‘burn-in’ are used to summarize inferences on the unknown parameters, where the sample mean or median is typically used as a point estimate of the parameter. In order to check for convergence, it is common to run multiple MCMC algorithms with different starting values, where each is called a chain, and inspect that the sampled posterior distributions are the same for the different chains. Another method to evaluate convergence is to use Gelman’s scale reduction statistic, $\hat{R}$, which has values near 1 for each parameter if the algorithm has converged (Gelman et al. 2003).

In fitting Bayesian models, MCMC algorithms are typically based on the Gibbs sampler (e.g. Casella and George 1992), which iteratively samples from the full conditional distribution for each parameter, conditioning on the current value of the other parameters. The full conditional distribution is the distribution for a parameter given the other parameters in the model. At iteration j , the Gibbs sampler for the SVCP model simulates successively from the following full conditional distributions:

$$\begin{aligned} \phi(j) &\sim [\phi|\boldsymbol{\beta}(j-1), \boldsymbol{\mu}_{\boldsymbol{\beta}}(j-1), \mathbf{T}(j-1), \tau^2(j-1), \mathbf{Y}] \\ \mathbf{T}(j) &\sim [\mathbf{T}|\boldsymbol{\beta}(j-1), \boldsymbol{\mu}_{\boldsymbol{\beta}}(j-1), \phi(j), \tau^2(j-1), \mathbf{Y}] \\ \tau^2(j) &\sim [\tau^2|\boldsymbol{\beta}(j-1), \boldsymbol{\mu}_{\boldsymbol{\beta}}(j-1), \mathbf{T}(j), \phi(j), \mathbf{Y}] \\ \boldsymbol{\mu}_{\boldsymbol{\beta}}(j) &\sim [\boldsymbol{\mu}_{\boldsymbol{\beta}}|\boldsymbol{\beta}(j-1), \phi(j), \mathbf{T}(j), \tau^2(j), \mathbf{Y}] \\ \boldsymbol{\beta}(j) &\sim [\boldsymbol{\beta}|\phi(j), \boldsymbol{\mu}_{\boldsymbol{\beta}}(j), \mathbf{T}(j), \tau^2(j), \mathbf{Y}]. \end{aligned} \quad (11)$$

Clearly, to use the Gibbs sampler, one must be able to derive the full conditional distribution for each parameter. The full conditional distribution for each parameter is derived by taking the product of the appropriate likelihood function and the priors for all the parameters and then simplifying the expression for the parameter of interest by ignoring terms that do not include the parameter of interest. The full conditional distributions using conjugate

priors are listed in the Appendix for $\mathbf{T}$, τ^2 , $\boldsymbol{\mu}_\beta$, and $\boldsymbol{\beta}$. Unlike the other parameters, the full conditional distribution of ϕ cannot be found in closed form.

When a conditional distribution for a parameter can only be calculated up to a normalizing constant, as is the case with ϕ , a Metropolis–Hastings (M–H) step can be used to draw a sample from the full conditional distribution of a parameter (e.g. Chib and Greenberg 1995). In the M–H step, we use a normal random walk proposal density that is centered on the current value of ϕ and has a variance s^2 that is tuned to produce an adequate acceptance rate of the proposed value of ϕ . The proposed value of ϕ is accepted if the ratio of the unnormalized full conditional distribution with the proposed value over the unnormalized full conditional distribution with the current value of ϕ is greater than 1 or, if not, is greater than a randomly drawn uniform variable with a range of (0,1). The unnormalized full conditional distribution for ϕ is provided in the Appendix.

An alternative method for simulating the spatial dependence parameter ϕ is slice sampling. Slice sampling does not require the proposal variance tuning of Metropolis–Hastings. Two useful slice sampling references are Neal (2003) and Agarwal and Gelfand (2005). Gelfand et al. (2003) use slice sampling to fit all the parameters associated with the SVCP model in Eq. 7, including ϕ . However, we instead sample from the full conditional distributions for the non-spatial dependence parameters because it is more efficient.

A feature of the SVCP model is that the correlation between explanatory variable coefficients is explicitly modeled. With the separable covariance matrix, the posterior correlation of the regression coefficients k and l across all locations is $T_{kl}/\sqrt{T_{kk}T_{ll}}$. Another expression for this correlation is

$$\frac{T_{kl}}{\sqrt{T_{kk}T_{ll}}} = \frac{\text{cov}(\beta_k(s), \beta_l(s+h))}{\sqrt{\text{cov}(\beta_k(s), \beta_k(s+h))\text{cov}(\beta_l(s), \beta_l(s+h))}}, \quad (12)$$

which does not depend on h , hence, the correlation of (β_k, β_l) does not depend on distance (h) between locations and is the same across the study area (Gelfand et al. 2004). We make use of this generalized coefficient correlation in the next section when evaluating dependence in the coefficients in the simulation studies.

3 Assessing marginal inference through simulation studies

In this section, we use two simulation studies to evaluate and compare the coverage probabilities and accuracy of the regression coefficients from the GWR and Bayesian SVCP models. We assess inferences on the coefficients both when there is no collinearity in the explanatory variables and when there is collinearity, expressed at various levels. The motivation for doing this is to explicitly test the assumption that the inferences on the coefficients from each model are valid, in the sense that the 95% confidence interval or credible

interval for each estimated coefficient contains the true value 95% of the time. The Bayesian 95% credible interval for a parameter is the range from the 0.025 quantile to the 0.975 quantile of the sampled posterior distribution. If the estimated coefficient intervals do not contain the true values more than 5% of the time, then there is clearly a problem with interpreting the coefficients, and the problem is more severe as the percent of intervals not containing the true values increases from 5%. We first evaluate the assumption of acceptable coverage when there is no collinearity because this is the most favorable, although unlikely, situation. We next evaluate the assumption of acceptable coverage when there is collinearity in the model and specify systematic increases in collinearity to inspect its effect on both the coverage probabilities and accuracy of the coefficients, as well as the strength of correlation in the estimated coefficients at each location and across the study area.

The model to generate the data for the two simulation studies is

$$y(s) = \beta_1^*(s)x_1(s) + \beta_2^*(s)x_2(s) + \varepsilon(s), \quad (13)$$

where the x_1 and x_2 are the first two principal components from a random sample drawn from a multivariate normal distribution of dimension ten with a mean vector of zeros and an identity covariance matrix, and the errors ε are sampled independently from a normal distribution with mean 0 and variance τ^{2*} . Note that there is no true intercept in the model used to generate the data and we do not fit an intercept in the simulation studies. The data points are equally spaced on a 10×10 grid, for a total of 100 observations. We use this size grid for computational considerations. The goal of the simulation studies is to use the model in Eq. 13 to generate the data and see if the regression coefficient estimates match β^* for both the GWR and SVCP models.

For each simulation study, we generate 200 realizations of the data process. In the first simulation study, the explanatory variables and the error terms are simulated for each realization of the data, while the true regression coefficients are fixed for all realizations. The true regression coefficients used to simulate the data are based not on a model, but explicitly on the coordinates of the data points in the study area. In the second simulation study, the true regression coefficients are simulated at each realization of the data, and thus vary from realization to realization. Another key difference between the two simulation studies is that there is no collinearity introduced into the explanatory variables in the first study and various levels of collinearity are utilized in the second simulation study. An additional contrast is that there is only one level of spatial dependence in the regression coefficients in the first simulation study, and there are three different levels of spatial dependence in the second simulation study.

3.1 Simulation study 1

The first simulation study has one fixed pattern of true coefficients that are used to generate the data, where there is strong spatial dependence in the

coefficients within each explanatory variable parameter. To achieve this, the true coefficient $\beta_1^*(s)$ is equal to the x coordinate at location s and the true coefficient $\beta_2^*(s)$ is equal to the y coordinate at location s . This results in a clear, increasing trend in each parameter across space, as shown in Fig. 1. The regression coefficients are fixed in this way to compare the GWR and SVCP models when the true coefficients are not based on either model, thereby eliminating any predisposition towards one of the models. This gives a baseline performance for the subsequent simulation study in this section. In this simulation study, two different error variances are used, $\tau^{2*} = 0.001$ and $\tau^{2*} = 1$, in two runs of the study, to evaluate the consistency of the results to the error level. The smaller value is used to effectively remove any complicating effect the error term may have in estimating the model parameters. In the SVCP model, simulation-based inference is carried out with 4,000 iterations in the MCMC routine, discarding the initial 2,000 iterations as the “burn-in”. Based on trace plots and Gelman’s $\hat{R}$ statistic (e.g. Gelman et al. 2003), the regression coefficients converged within 2,000 iterations of the Gibbs sampler. We verified that the number of MCMC iterations was adequate by performing an experiment estimating the SVCP model parameters with a number of realizations of the data using a number of iterations that was one order of magnitude larger than in the simulation study for each realization. We found no substantial differences in the measures of accuracy and reliability of the regression coefficients. Also concerning the MCMC implementation, we do not perform thinning on the MCMC samples because it is not necessary in this case, where we are interested in functional estimates of the model parameters. See MacEachern and Berliner (1994) for a justification of not thinning, or subsampling, MCMC samples. We did, however, test the effect of thinning by running an experiment estimating the SVCP model parameters with for a number of realizations with and without thinning, and found no difference in the measures of accuracy and reliability of interest for the regression coefficients with subsampling. After running the MCMC algorithm to estimate the

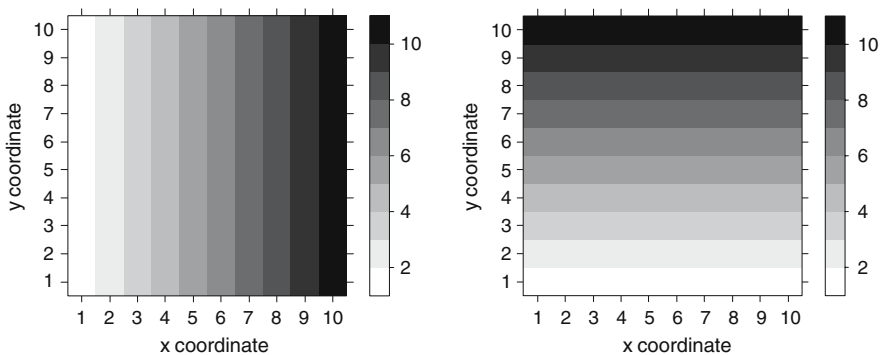


Fig. 1 The gray-scale shading shows the coefficient pattern for each β^* parameter in simulation study 1. The left plot is for β_1^* and the right plot is for β_2^* . The parameter values range from 1 (lightest) to 10 (darkest)

parameters in the Bayesian SVCP model, the corresponding GWR model is also fitted to the same data, using cross-validation to estimate the kernel bandwidth.

The prior specification for the SVCP model for this simulation study is as follows. We use a vague normal, $N(\mathbf{0}, 10^4 \mathbf{I})$, for $\boldsymbol{\mu}_\beta$, a three-dimensional inverse Wishart, $IW(3, 0.1\mathbf{I})$, for $\mathbf{T}$, and an inverse gamma, $IG(2, 1)$, for τ^2 , where $\mathbf{I}$ is the identity matrix of dimension p . For the spatial dependence parameter ϕ , we use a gamma, $G(0.15, 0.05)$. The hyperparameters for this gamma prior are chosen so that the prior mean approximately solves the spatial correlation function set equal to 0.05, where the spatial range is set to half the maximum inter-point distance from the distance matrix $\mathbf{D}$. The spatial range is the distance beyond which the spatial association becomes negligible, and is approximately 3ϕ with our parameterization of the covariance function. The spatial decay parameter can be weakly identifiable with the SVCP model if the data do not exhibit strong spatial dependence, so an informative prior is used for ϕ . A distribution that sets the prior mean spatial range to roughly half that of the maximum pairwise distance usually results in stable MCMC behavior (Banerjee and Johnson 2005). The calculation for the mean is found from solving $\exp(-(1/2 \cdot \max(\mathbf{D}))/\phi) = 0.05$ for ϕ .

In order to ultimately perform inference on the parameters in the SVCP model, starting values are required for each of the parameter estimates in the MCMC. The starting values for each SVCP model parameter estimate in each realization are assigned random values from certain distributions. The spatial dependence value is drawn from a uniform distribution over the range of pairwise distances in the study area. The coefficient means are drawn randomly from a multivariate normal distribution with zero means and the identity covariance matrix. To ensure that the starting value for the coefficient covariance matrix $\mathbf{T}$ is positive definite, a matrix is sampled from a multivariate normal distribution with a mean of the identity matrix and a diagonal matrix of 0.00001 for the covariance matrix. This gives a starting value for $\mathbf{T}$ that is close to the identity matrix.

To evaluate the coverage and accuracy of the regression coefficients, we calculate numerous summary statistics. For each realization of the process, we calculate the 95% credible intervals for each coefficient in the SVCP model and the 95% confidence intervals for each GWR coefficient. The confidence intervals for GWR are calculated in the standard way, using the standard errors derived from Eq. 5. To calculate the 95% coverage probabilities, the true coefficients are compared to the 95% intervals obtained for the respective coefficient in each data realization and then the total number of realizations that contain the true values are summed and divided by the number of realizations. In order to create a summary measure for all the realizations that is straightforward to present in a tabular format, the means of the coverage probabilities are calculated for each explanatory variable from the coefficients corresponding each the variable. The accuracy of the regression coefficients is measured by calculating the root mean square error (RMSE) of all coefficients at each realization. The RMSE is the square root of the mean of the squared

deviations of the estimates from the true values. The MSE is equal to the sum of the bias² and variance of an estimate, and it should be small for an accurate estimator. The average RMSE for all realizations is then calculated by averaging the RMSE's from all of the individual realizations.

The mean RMSE of the regression coefficients and the 95% coverage probabilities for each explanatory variable coefficient in simulation study 1 are listed in Table 1 for the Bayesian SVCP model and the GWR model for $\tau^{2*} = 0.001$ and in Table 2 for $\tau^{2*} = 1$. The results show that the Bayesian regression model has substantially lower RMSE for the coefficients than does the GWR model. In fact, the GWR RMSE is three times that of the SVCP RMSE with the smaller error variance. The RMSE is 30% higher in the GWR model than in the SVCP model with the larger error variance. The Bayesian regression model has nearly 100% coverage (rounded to the nearest integer) of the true regression coefficients in the 95% confidence intervals, while the GWR model has only about 63% coverage of the true coefficients with the smaller error variance. The coverage probabilities decrease to 98% for the SVCP model and increase to 82% for GWR with the larger error variance. Ideally, the coverage probabilities would be 95% for all coefficients using the 95% credible intervals for the Bayesian model and the 95% confidence intervals for the GWR model. There is evidence that the SVCP model is over covering the true parameters by overestimating the uncertainty. For the GWR model, the results with the smaller error variance mean that, on average, 37% of the coefficient confidence intervals do not contain the true coefficient values. This is clearly an unfavorable result for the GWR model, and if the result generalizes, casts significant doubt on inferences from this model. The results for this simulation study are from only one level of spatial dependence in the coefficients. To summarize the results from this simulation study, the Bayesian model outperforms the GWR model in terms of both regression coefficient coverage of the true values and accuracy when the true coefficients are not based on any model, there is strong spatial dependence in the regression coefficients, and there is no substantial collinearity in the explanatory variables.

3.2 Simulation study 2

In the second simulation study, the true coefficients used to simulate the data are simulated from the fixed, true parameter values for the SVCP model. We fix the values for the parameters in the Bayesian model, other than the

Table 1 Average root mean square error (RMSE) and percent coverage (CP) of the 95% confidence intervals of the regression coefficients with GWR and percent coverage of the 95% credible intervals with the SVCP model in simulation study 1 with true error variance = 0.001

Method	RMSE (β)	Mean CP (β_1)	Mean CP (β_2)
Bayesian	0.138	100%	100%
GWR	0.414	63%	63%

Table 2 Average root mean square error (RMSE) and percent coverage (CP) of the 95% confidence intervals of the regression coefficients with GWR and percent coverage of the 95% credible intervals with the SVCP model in simulation study 1 with true error variance = 1

Method	RMSE (β)	Mean CP (β_1)	Mean CP (β_2)
Bayesian	0.452	98%	98%
GWR	0.595	82%	82%

regression coefficients, and simulate the true coefficients using the multivariate normal distribution and the fixed values for the coefficient means, covariances, and spatial dependence parameter. The coefficient covariance matrix at all locations is set to $\mathbf{T}^* = \begin{pmatrix} 0.1 & 0.0 \\ 0.0 & 0.5 \end{pmatrix}$ and the coefficient means are set to $\boldsymbol{\mu}_\beta^* = (1, 5)$. The spatial dependence parameter is first set to $\phi^* = 10$, then $\phi^* = 5$, and finally $\phi^* = 1$ to make three different sets of 200 realizations of the coefficient process. This is done to evaluate the performance of the GWR and SVCP model under three different levels of spatial dependence in the regression coefficients. We use 2,000 iterations in the MCMC routine for simulation-based inference of the SVCP model parameters, with a “burn-in” of 1,000 iterations, and also fit the GWR model to the same data. Based on trace plots and Gelman’s $\hat{R}$ statistics, the regression coefficients converged for individual realizations of the coefficient process within 1,000 iterations of the Gibbs sampler. As in simulation study 1, we performed an experiment to assess the sensitivity to the number of MCMC iterations and subsampling, and found no substantial differences in the measures of accuracy and reliability of interest for the regression coefficients with either subsampling or with the larger magnitude of iterations. For this experiment, we used 200 realizations of the coefficient process with one true value of the spatial dependence parameter, $\phi^* = 10$, and no dependence in the explanatory variables.

The prior specification for the SVCP model for this simulation study is as follows. We use a vague normal, $N(\mathbf{0}, 10^4 \mathbf{I})$, for $\boldsymbol{\mu}_\beta$, a three-dimensional inverse Wishart, $IW(3, 0.1\mathbf{I})$, for $\mathbf{T}$, and an inverse gamma, $IG(2, 1)$, for τ^2 , where $\mathbf{I}$ is the identity matrix of dimension p . For the spatial dependence parameter ϕ , we use a gamma, $G(1.5, 0.5)$, which, as in simulation study 1, is an informative prior with a prior spatial range of approximately half the study area. The same starting value distributions for the SVCP model parameter estimates are used as in simulation study 1.

Examples of the spatial pattern in individual realizations of the coefficient process are shown in Fig. 2 for $\phi^* = 1$, Fig. 3 for $\phi^* = 5$, and Fig. 4 for $\phi^* = 10$. The figures show patterns in the coefficients for each variable, but the patterns are less clear and dramatic than those for the true coefficients in the first simulation study displayed in Fig. 1, where there is more spatial dependence. In general, as ϕ^* increases there are more consistent and clear patterns in the true regression coefficients, and more global variation in the coefficients. Conversely, there is more local variation with smaller ϕ^* , resulting in

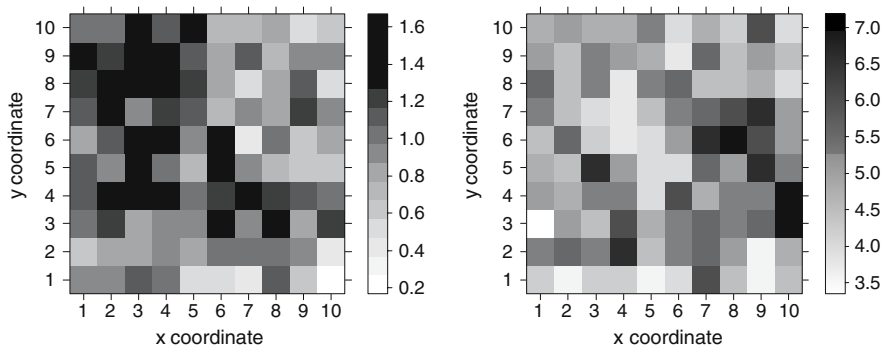


Fig. 2 The *gray-scale shading* shows the coefficient pattern for each β^* parameter for one coefficient realization in simulation study 2 when $\phi^* = 1$. The *left* plot is for β_1^* and the *right* plot is for β_2^*

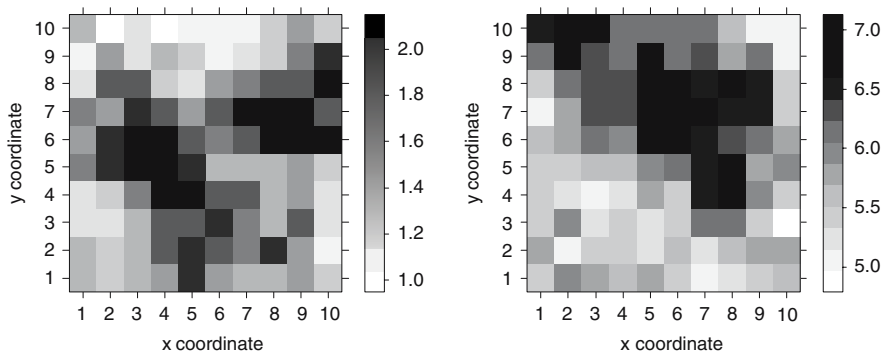


Fig. 3 The *gray-scale shading* shows the coefficient pattern for each β^* parameter for one coefficient realization in simulation study 2 when $\phi^* = 5$. The *left* plot is for β_1^* and the *right* plot is for β_2^*

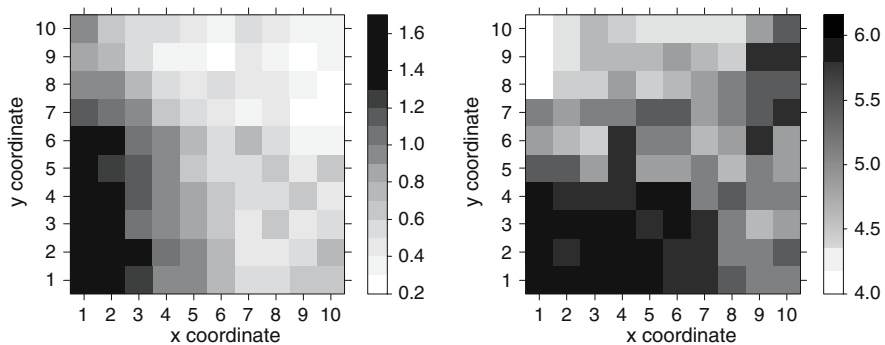


Fig. 4 The *gray-scale shading* shows the coefficient pattern for each β^* parameter for one coefficient realization in simulation study 2 when $\phi^* = 10$. The *left* plot is for β_1^* and the *right* plot is for β_2^*

less smooth patterns in the coefficients. For this simulation study, we start with no substantial collinearity in the model and systematically increase it until the explanatory variables are nearly perfectly collinear. This is done by replacing one of the original explanatory variables with one created from a weighted linear combination of the original explanatory variables, where the weight determines the amount of correlation of the variables. The formula for the new weighted variable is

$$x_2^c = c \cdot x_1 + (1 - c) \cdot x_2, \quad (14)$$

where x_2^c replaces x_2 in the model in Eq. 13 and c is the weight between 0 and 1.

To evaluate the coverage and accuracy of the regression coefficients from the GWR and SVCP model, we use the same coverage probability and RMSE calculations described for simulation study 1. In addition, we calculate the correlation in the estimated regression coefficients from both models. There are two types of correlation of interest in the estimate regression coefficients. One is the overall correlation coefficient (C_{12}) between the sets of estimated coefficients for two explanatory variables, and the other is the estimated local correlation coefficient (C_{12}^s) between two coefficients at any location s . The local coefficient correlation at each location is given by Eq. 12 for the SVCP model. The local coefficient correlation in the GWR model is calculated using the previously defined Eq. (6). The overall correlation between sets of coefficients is calculated for the SVCP and GWR models using previously defined Eq. (4).

The results of this simulation study are listed in Table 3 for the Bayesian SVCP model and Table 4 for the GWR model. The tables show the mean statistics calculated from the 200 realizations of the coefficient process. The coverage probabilities generally increase in both the SVCP and GWR models as ϕ^* increases. The tables also show that the SVCP model has substantially larger mean coverage probabilities than does GWR when there is no collinearity in the data ($c = 0$), regardless of the level of ϕ^* . As the collinearity in the explanatory variables increases, the coverage probabilities vary little in the SVCP model and increase markedly in the GWR model. The coverage probabilities increase with increasing collinearity in the GWR model because the variances of the coefficients increase as well, as shown in the columns of coefficient variances in Table 4. It is a well-known result in statistics that regression coefficient variances increase in the presence of collinearity (Neter et al. 1996). The coverage probability increases with decreasing estimator bias and increases with increasing estimator variance. Therefore, looking only at the coverage probabilities for the GWR model is misleading because the increase in coverage probabilities is due directly to increasingly imprecise estimates. Figures 5 and 6 show the coverage probabilities for the regression coefficients in the GWR and SVCP models, respectively, when there is no correlation in the explanatory variables and $\phi^* = 10$. The plots show that there is more variation in the coverage probabilities with the GWR model than with the SVCP model.

Table 3 Results of simulation study 2 for the Bayesian SVCP model

Corr (X)	Mean CP (β_1)	Mean CP (β_2)	C_{12}	C^s_{12}	RMSE (β)	RMSE (y)
$\phi^* = 1$						
0.000	0.91	0.86	0.02	0.01	0.657	0.904
0.127	0.92	0.84	0.11	0.01	0.664	0.923
0.442	0.91	0.82	0.29	0.10	0.681	0.947
0.755	0.93	0.80	0.54	0.15	0.706	0.958
0.937	0.95	0.84	0.73	0.18	0.780	0.947
0.995	0.96	0.93	0.86	0.22	1.382	0.920
$\phi^* = 5$						
0.000	0.95	0.92	0.01	0.01	0.458	0.889
0.127	0.95	0.91	0.09	0.03	0.465	0.900
0.442	0.96	0.89	0.24	0.06	0.486	0.912
0.755	0.96	0.88	0.52	0.12	0.522	0.922
0.937	0.96	0.89	0.70	0.12	0.610	0.920
0.995	0.94	0.93	0.85	0.16	1.221	0.911
$\phi^* = 10$						
0.000	0.97	0.93	0.11	0.06	0.378	0.892
0.127	0.97	0.93	0.17	0.06	0.384	0.903
0.442	0.97	0.92	0.30	0.06	0.407	0.910
0.755	0.97	0.91	0.51	0.07	0.440	0.919
0.937	0.97	0.92	0.66	0.05	0.526	0.920
0.995	0.94	0.94	0.90	0.10	1.079	0.914

The columns listed in order are the explanatory variable correlation, mean coverage probabilities for each explanatory variable coefficient, the mean overall correlation between the variable coefficients, the mean local coefficient correlation at each location, the average root mean square error (RMSE) of the coefficients, and the RMSE of the response

The RMSE is a better measure of parameter estimate accuracy in the presence of collinearity because it increases with increasing bias and increasing estimator variance. Figure 7 shows the average RMSE values for the regression coefficients for the SVCP and GWR models at varying levels of collinearity for the three levels of ϕ^* . The average RMSE values indicate that the coefficients from the SVCP model are closer to their true values than are those from GWR at each level of collinearity and level of spatial dependence in the true regression coefficients. Not surprisingly, the average RMSE increases consistently with increasing collinearity for both models, showing that the estimated coefficients move away from their true values. Intuitively, the average RMSE values for both the GWR and SVCP models decrease as the spatial dependence increases because the models are more appropriate for data with more spatial dependence. The results show that the GWR model coefficients in general are more biased than the SVCP model coefficients because the average RMSE for the SVCP model is consistently less than for GWR and the coverage probabilities are larger for the SVCP model than for GWR.

The mean overall and local correlations calculated from all realizations in the SVCP and GWR models are listed in Tables 3 and 4. The results show that the Bayesian model better controls the correlation in the regression

Table 4 Results of simulation study 2 for the GWR model

Corr (X)	Mean CP (β_1)	Mean CP (β_2)	C_{12}	C_{12}^s	Var (β_1)	Var (β_2)	RMSE (β)	RMSE (y)
$\phi^* = 1$								
0.000	0.59	0.36	0.08	0.01	0.03	0.03	0.718	1.134
0.127	0.59	0.39	0.00	-0.10	0.03	0.04	0.719	1.102
0.442	0.59	0.43	-0.24	-0.40	0.03	0.05	0.730	1.082
0.755	0.68	0.54	-0.56	-0.72	0.05	0.10	0.781	1.062
0.937	0.81	0.71	-0.83	-0.92	0.17	0.30	0.989	1.007
0.995	0.89	0.87	-0.98	-0.99	2.50	3.04	2.471	1.113
$\phi^* = 5$								
0.000	0.78	0.60	0.10	0.02	0.04	0.04	0.486	0.906
0.127	0.78	0.61	0.04	-0.10	0.04	0.05	0.487	0.908
0.442	0.78	0.66	-0.15	-0.39	0.04	0.06	0.509	0.917
0.755	0.82	0.74	-0.44	-0.71	0.06	0.12	0.581	0.922
0.937	0.89	0.84	-0.77	-0.92	0.19	0.33	0.836	0.925
0.995	0.91	0.91	-0.98	-0.99	2.96	3.60	2.593	0.913
$\phi^* = 10$								
0.000	0.80	0.63	0.07	0.02	0.03	0.03	0.400	0.907
0.127	0.80	0.65	0.01	-0.10	0.03	0.04	0.404	0.909
0.442	0.80	0.70	-0.17	-0.39	0.03	0.05	0.428	0.914
0.755	0.84	0.77	-0.45	-0.72	0.05	0.10	0.494	0.920
0.937	0.90	0.86	-0.77	-0.92	0.16	0.28	0.718	0.923
0.995	0.92	0.92	-0.98	-0.99	2.40	2.93	2.239	0.912

The columns correspond to the explanatory variable correlation, mean coverage probabilities for each explanatory variable coefficient, the mean overall correlation between the variable coefficients, the mean local coefficient correlation at each location, the variance of each explanatory variable coefficient, the RMSE of the coefficients, and the RMSE of the response

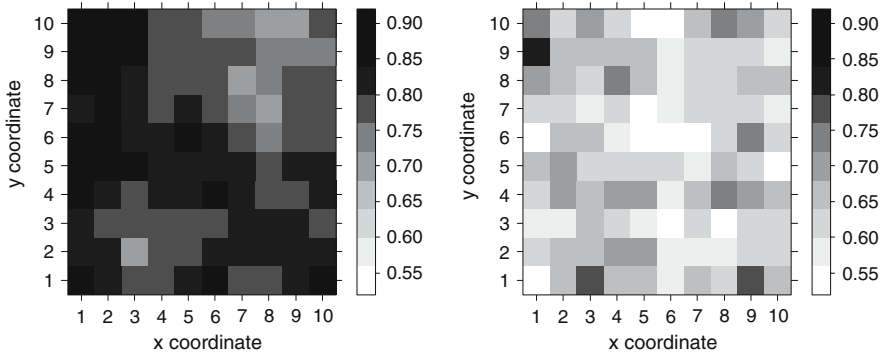


Fig. 5 The gray-scale shading shows the mean coverage probabilities for each β^* parameter when $\phi^* = 10$ in simulation study 2 for the GWR model. The left plot is for β_1^* and the right plot is for β_2^*

coefficients than does the GWR model. The GWR coefficients become almost perfectly negatively correlated overall with very strong explanatory variable collinearity, while the SVCP model coefficients demonstrate strong levels of positive overall correlation. Similarly, the mean local correlation of

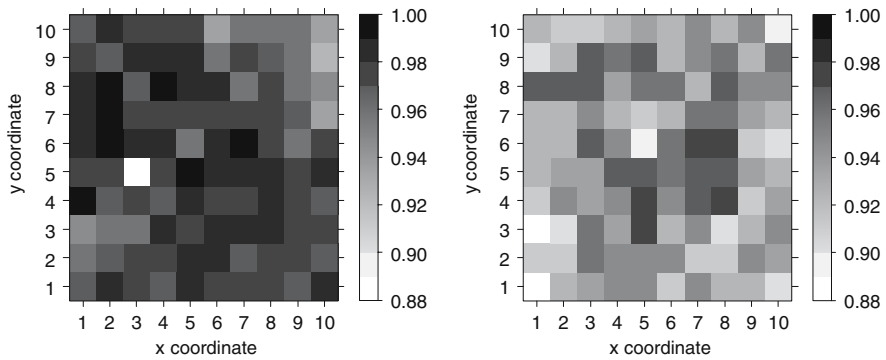


Fig. 6 The *gray-scale shading* shows the mean coverage probabilities for each β^* parameter when $\phi^* = 10$ in simulation study 2 for the SVCP model. The *left* plot is for β_1^* and the *right* plot is for β_2^*

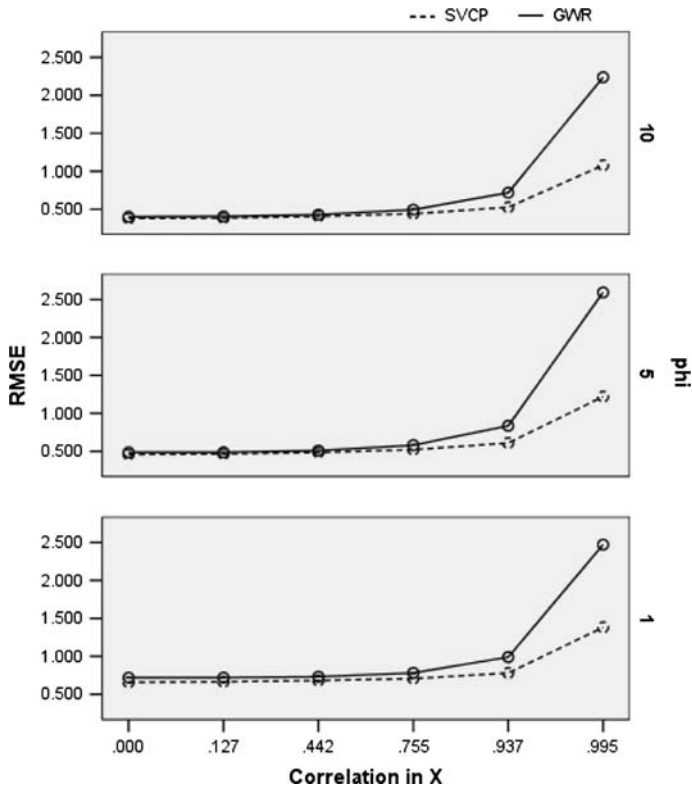


Fig. 7 The *lines* show the mean RMSE for regression coefficients from the SVCP (*dashed*) and the GWR (*solid*) models at specific levels of collinearity in the explanatory variables and at different levels of ϕ^* in simulation study 2

the coefficients is strongly negative with strong collinearity in the GWR model, while the mean local correlation is weakly positive with the SVCP model. Furthermore, the SVCP model appears self-tuning with regard to the level of collinearity in the model and its subsequent effect on the estimated coefficients. The variances of the coefficients do not increase dramatically with increasing collinearity in the SVCP model as they do in the GWR model, as is evident from the relatively consistent coverage probabilities in the SVCP model.

4 Conclusions

There has been an increasing interest in spatially varying relationships between variables in recent years in both the statistics and geography literature. Recent attempts at modeling these relationships have resulted in geographically weighted regression and Bayesian regression models with spatially varying coefficient processes. While these models have been applied to numerous real world datasets in the literature, there has been a conspicuous lack of attention to the validation of marginal inferences derived from these models. Our work uses a series of simulation studies to evaluate the accuracy of the regression coefficients for both the Bayesian SVCP and GWR models under the presence of collinearity. While a simulation study does not prove a general result, it does demonstrate that marginal inferences can potentially be misleading. There is additional complexity in implementing the Bayesian regression model compared to the GWR model; however, the simulation study results in this paper help to justify the benefits of the additional complexity of the Bayesian model. The Bayesian regression model provides more accurate regression coefficient estimates than does GWR in both the absence and presence of explanatory variable collinearity and produces less correlated coefficients in the presence of moderate and strong positive variable correlation. It is not unexpected that the Bayesian SVCP model accommodates collinearity better than GWR does given previous work which shows that ridge regression coefficient estimates may be viewed as Bayesian regression coefficient estimates under specific vague priors (Goldstein 1976). Wheeler and Calder (2006) show the Bayesian SVCP model regression coefficients as ridge regression coefficients, which are constrained in size to counter increased estimator variance due to the presence of collinearity in the model. In addition, the Bayesian SVCP model provides the posterior distribution for each parameter through the iteratively drawn samples, while the GWR model only gives a point estimate and corresponding standard error for each parameter. Moreover, Bayesian regression handles spatial dependence in the data through a single statistical model, whereas GWR is essentially an ensemble of separate models using shared data. In summary, the Bayesian regression model offers the spatial analyst more flexibility in modeling spatially varying relationships, and produces more interpretable and accurate inferences than does GWR.

One argument against using Bayesian models has been the difficulty in implementing them in practice. However, more general-purpose software for Bayesian inference is becoming available. At the forefront of the Bayesian software movement is WinBUGS (<http://www.mrc-bsu.cam.ac.uk/bugs/winbugs/contents.shtml>), which uses Gibbs sampling and other simulation methods for inference in a wide variety of Bayesian models. GeoBUGS is available in WinBUGS and can be utilized for Bayesian spatial models commonly found in epidemiological studies and other spatial applications. There are also MCMC packages available for R. All of the packages mentioned are free and available on the Internet. Inconveniently, GeoBUGS cannot currently perform Bayesian inference on the jointly specified SVCP model used in this paper, but one can imagine that future versions will overcome this limitation, as the program can currently handle the conditionally specified SVCP model. Regardless, we hope the exposition of the SVCP model provided in this paper makes the implementation of this type of Bayesian model more apparent and provides a practical guide to an alternative to GWR for spatially varying coefficient regression.

Appendix

As mentioned in the paper, the posterior distribution of a parameter is proportional to the product of the likelihood of the data given the parameter and the priors for the parameters. The posterior can be expressed as the full conditional up to a normalizing constant. The full conditional for a parameter is the distribution for a parameter given the other model parameters. If the distribution of a full conditional for a parameter is recognizable, one can sample directly from it in a Gibbs sampler. However, if the distribution of the full conditional is not recognizable, one can sample from it using a Metropolis–Hastings algorithm or slice sampling.

In order to perform inference on the model parameters, we must write the posterior distribution for each unknown parameter using the likelihood. The derivation of the full conditional distributions in this paper utilizes two versions of the likelihood. The likelihood for the SVCP model with $\mathbf{Y}$ as defined in Eq. 7 of the paper text is

$$L(\boldsymbol{\mu}_\beta, \beta, \tau^2, \phi, \mathbf{T}; \mathbf{y}) = |\tau^2 \mathbf{I}|^{-1/2} \times \exp \left\{ -\frac{1}{2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\tau^2 \mathbf{I})^{-1} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) \right\}. \quad (1)$$

We can integrate this likelihood with respect to $\boldsymbol{\beta}$ to reduce the autocorrelation in the Markov chain. The likelihood with integrating over $\boldsymbol{\beta}$ is

$$L^*(\boldsymbol{\mu}_\beta, \tau^2, \phi, \mathbf{T}; \mathbf{y}) = |\boldsymbol{\Psi}|^{-1/2} \times \exp \left\{ -\frac{1}{2} (\mathbf{y} - \mathbf{X}^* \boldsymbol{\mu}_\beta)^T (\boldsymbol{\Psi})^{-1} (\mathbf{y} - \mathbf{X}^* \boldsymbol{\mu}_\beta) \right\}, \quad (2)$$

where $\boldsymbol{\Psi} = (\mathbf{X}(\mathbf{H}(\phi) \otimes \mathbf{T})\mathbf{X}^T + \tau^2 \mathbf{I})$ and $\mathbf{X}^*$ is the $n \times p$ matrix of covariates. As an example of the use of both likelihoods, the full conditional distribution

for μ_{β} is derived using the likelihood integrated over β, L^* , and the full conditional distribution for τ^2 is derived using the likelihood L .

We first derive the full conditionals that are recognizable distributions using the likelihood and priors. The full conditional for the error variance is

$$[\tau^2 | \beta; \mathbf{y}] \sim L \times p(\tau^2) = IG\left(a + n/2, b + \frac{1}{2}(\mathbf{y} - \mathbf{X}\beta)^T(\mathbf{y} - \mathbf{X}\beta)\right). \quad (3)$$

The full conditional for the coefficient covariance matrix at any location is

$$\begin{aligned} [\mathbf{T} | \mu_{\beta}, \beta, \phi; \mathbf{y}] &\sim L \times p(\mathbf{T}) \times p(\beta) \\ &= IW\left(v + n, \sum_i \sum_j (\mathbf{H}^{-1}(\phi))_{ij} (\beta(s_j) - \mu_{\beta})(\beta(s_i) - \mu_{\beta})^T + \mathbf{\Omega}\right), \end{aligned} \quad (4)$$

where $\beta = (\beta(s_1), \beta(s_2), \dots, \beta(s_n))^T$ and $\mu_{\beta} = (\mu_{\beta_1}, \mu_{\beta_2}, \dots, \mu_{\beta_p})^T$. The full conditional for the coefficient means is

$$[\mu_{\beta} | \mathbf{T}, \tau^2, \phi; \mathbf{y}] \sim L^* \times p(\mu_{\beta}) = N(\mathbf{m}, \mathbf{S}), \quad (5)$$

where $\mathbf{S} = [(\sigma^2 \mathbf{I})^{-1} + \mathbf{X}^{*T} \mathbf{\Psi}^{-1} \mathbf{X}^*]^{-1}$ and $\mathbf{m} = \mathbf{S}(\mathbf{X}^{*T} \mathbf{\Psi}^{-1} \mathbf{y} + (\sigma^2 \mathbf{I})^{-1} \mu)$. The full conditional distribution for β is

$$[\beta | \mu_{\beta}, \phi, \mathbf{T}, \tau^2; \mathbf{y}] \sim L \times p(\beta) = N(\mathbf{AC}, \mathbf{A}), \quad (6)$$

where $\mathbf{A} = (\mathbf{X}^T \mathbf{X} / \tau^2 + \mathbf{H}^{-1}(\phi) \otimes \mathbf{T}^{-1})^{-1}$ and $\mathbf{C} = \mathbf{X}^T \mathbf{y} / \tau^2 + (\mathbf{H}^{-1}(\phi) \otimes \mathbf{T}^{-1})(\mathbf{1} \otimes \mu_{\beta})$.

The density of the unnormalized full conditional of ϕ is not a recognizable distribution and is

$$\begin{aligned} p(\phi | \mathbf{T}, \beta, \mu_{\beta}; \mathbf{y}) &\propto L \times p(\phi) \times p(\beta) \sim |\mathbf{H}(\phi) \otimes \mathbf{T}|^{-1/2} \\ &\times \exp\left\{-\frac{1}{2}(\beta - (\mathbf{1} \otimes \mu_{\beta}))^T (\mathbf{H}(\phi) \otimes \mathbf{T})^{-1} (\beta - (\mathbf{1} \otimes \mu_{\beta}))\right\} \\ &\times \phi^{z-1} \exp(-\lambda\phi). \end{aligned} \quad (7)$$

This density is used in the Metropolis–Hastings algorithm to estimate ϕ .

The parameterization of the gamma distribution used in this paper is

$$[\phi] \propto \phi^{z-1} \exp(-\lambda\phi), \quad (8)$$

and the parameterization of the inverse Wishart distribution used in this paper is

$$[\mathbf{T}] \propto |\mathbf{T}|^{-(v+p+1)/2} \exp\left(-\frac{1}{2} \text{trace} \mathbf{\Omega} \mathbf{T}^{-1}\right). \quad (9)$$

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